CS 418: Introduction to Data Science

Final Project Report - Group #15

Predicting Red Wine Quality: Regression & Classification

A Physicochemical Approach using Machine Learning

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1. Introduction

Red wine quality assessment has traditionally relied on expert sensory evaluation, a process that is subjective, time-consuming, and difficult to scale. Physicochemical analysis offers an objective, reproducible alternative: by measuring chemical properties of a wine sample, it may be possible to predict a quality rating that aligns with expert judgment.

This project investigates whether machine learning models trained on physicochemical measurements can reliably predict the quality of Portuguese Vinho Verde red wine. We address the problem as two complementary tasks: (1) regression: predicting the exact numerical quality score; and (2) binary classification: determining whether a wine is "good" (quality score >= 7) or "not good" (score < 7).

The dataset used is the UCI Red Wine Quality dataset, which contains 1599 wine samples described by 11 physicochemical features and a quality score assigned by wine experts on a scale from 0 to 10. We evaluate multiple models and feature sets for each task, report quantitative performance metrics, and draw conclusions about which chemical properties are most predictive of wine quality.

2. Data Preparation

2.1 Dataset Overview

The dataset was loaded from winequality-red.csv (semicolon-seperated). It contains 1599 observations and 12 columns: 11 continuous physicochemical input features and one integer target variable (quality).

Property	Value
Total samples	1,599
Input features	11 (all float64)
Target variable	quality (int64, range 3-8)
Missing values	None

2.2 Summary Statistics

Key descriptive statistics for all features are shown in the table below. Notable observations: alcohol ranges from 8.4% to 14.9% (mean 10.4%); volatile acidity ranges 0.12 - 1.58 g/L; quality scores range 3 - 8 with a mean of approximately 5.64.

Feature	Mean	Std	Min	Max
fixed acidity	8.32	1.74	4.60	15.90
volatile acidity	0.528	0.179	0.120	1.580
citric acid	0.271	0.195	0.000	1.000
residual sugar	2.539	1.410	0.900	15.500
chlorides	0.0875	0.0471	0.012	0.611
free sulfur dioxide	15.87	10.46	1.00	72.00
total sulfur dioxide	46.47	32.90	6.00	289.00
density	0.9967	0.00189	0.9901	1.0037
pH	3.311	0.154	2.740	4.010
sulphates	0.658	0.170	0.330	2.000
alcohol	10.42	1.07	8.40	14.90
quality	5.636	0.808	3.00	8.00

2.3 Missing Value Check

A full missing value check confirmed that all 1,599 rows are complete across all 12 columns. No imputation or row removal was required.

2.4 Outlier Detection

Outliers were identified using the 1.5xIQR rule for each physicochemical feature. All features contain some flagged outliers, with total sulfur dioxide (55 flagged), residual sugar (46 flagged), and chlorides (87 flagged) having the most. Rather than removing these observations, which could reflect genuine extremes in wine composition, all rows were retained to preserve the full distribution of the data.

2.5 Feature Engineering

A binary target variable quality_label was engineered from the quality score: wines with quality >= 7 were labeled "Good" (1) and all others "Not Good" (0). This variable would be used as the classification target.

Class	Count	Proportion
0: Not Good (<7)	1,382	86.4%
1: Good (>=7)	217	13.6%

The class imbalance (approximately 86.4% compared to 13.6%) is an important consideration for classification, as models can achieve high accuracy simply by predicting "not good" for every sample.

2.6 Train/Test Split

The data was split 80/20 (train/test) with stratification on quality_label to preserve the class ratio in both sets. All features were standardized using StandardScaler fitted on the training set only.

Split	Samples	Good Rate
Training set	1,279	13.6%
Test set	320	13.4%

3. Exploratory Data Analysis

3.1 Quality Score Distribution

The quality scores follow a roughly normal distribution centered around 5 and 6, with very few wines scoring 3 or 8. This concentration in the middle scores reflects the difficulty of distinguishing subtle quality differences and contributes to the moderate performance of regression models.

Figure 1: Distribution of quality scores across all 1,599 wine samples.

3.2 Correlation Analysis

The correlation heatmap below reveals the pairwise linear relationships between all features and the quality target. Alcohol content has the strongest positive correlation with quality (r = 0.48), while volatile acidity has the strongest negative correlation (r = -0.39). Several features are also correlated with each other: notably fixed acidity, citric acid, and density - which has implications for model design.

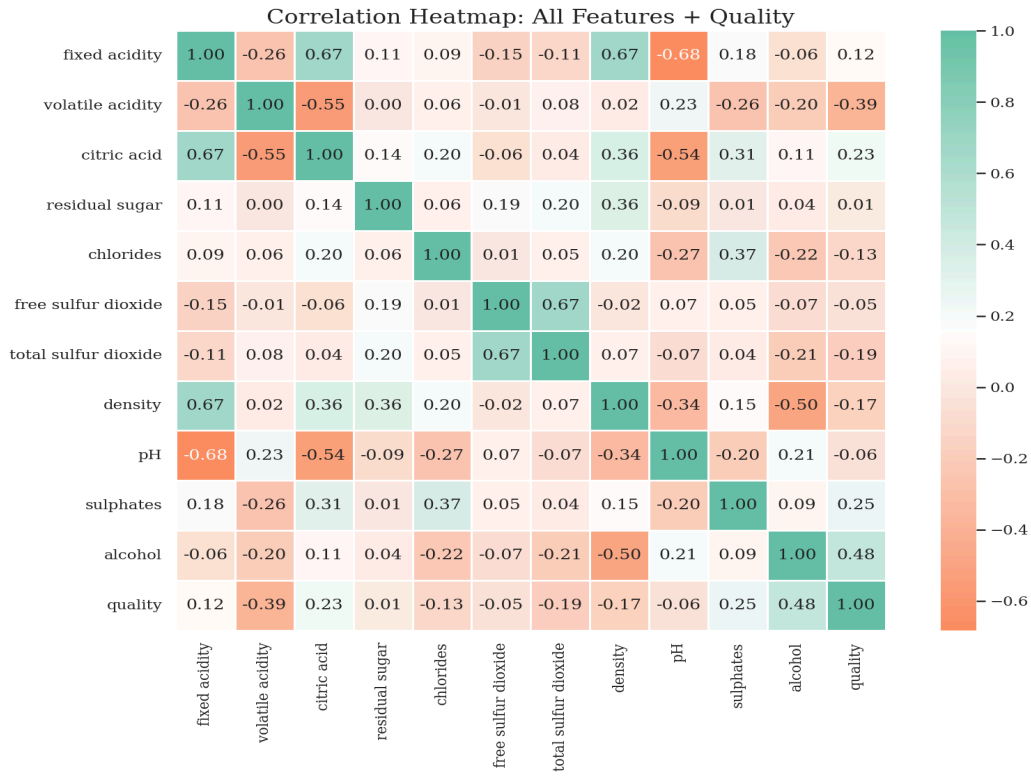


Figure 2: Correlation heatmap of all features and quality score.

3.3 Feature Importance by Correlation

The following charts rank features by their absolute correlation with quality and show the direction of each relationship. Alcohol and volatile acidity are the two most informative features, followed by sulphates and citric acid. Residual sugar, free sulfur dioxide, and pH show very weak correlations with quality.

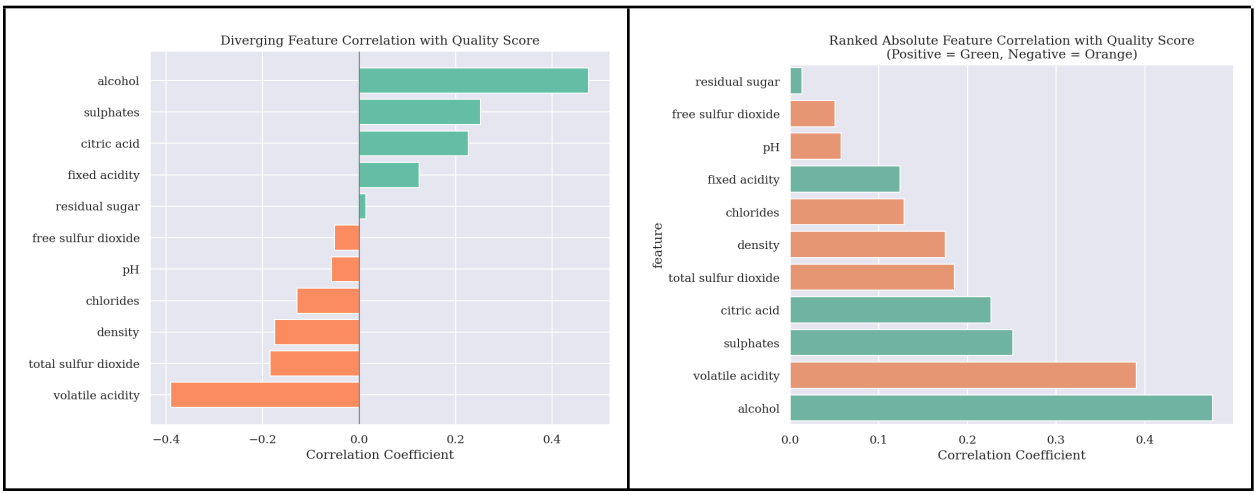


Figure 3: Diverging feature correlations with quality score (green = positive, orange = negative).
Figure 4: Ranked absolute feature correlations with quality score.

3.4 Alcohol Distribution by Quality Label

Good wines (quality ≥ 7) consistently show higher alcohol content than not-good wines. The histogram below illustrates this separation clearly, confirming that alcohol is the strongest single predictor of wine quality in this dataset.

3.5 Scatter Plot: Alcohol vs Volatile Acidity

A scatter plot of the two highest-correlated features shows partial but imperfect class separation. Good wines cluster toward higher alcohol and lower volatile acidity, while not-good wines are more dispersed. The overlap between classes explains why no single feature or simple linear boundary can perfectly classify wine quality.

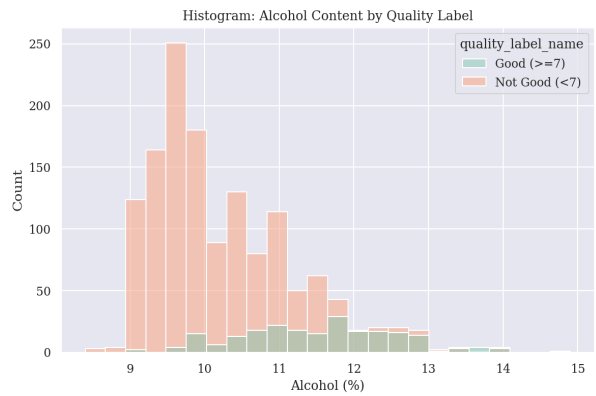


Figure 5: Histogram of alcohol by quality label.

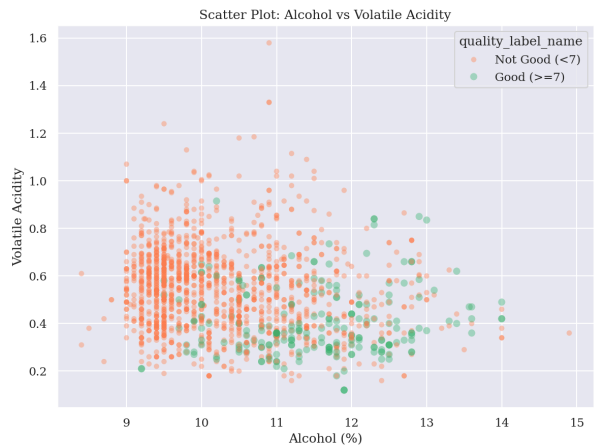


Figure 6: Scatter plot of alcohol vs. volatile acidity, colored by quality label.

3.6 Box Plots: Features by Quality Label

Box plots for all 11 features broken down by quality label confirm the patterns identified in the correlation analysis. Good wines show notably **higher alcohol**, **higher sulphates**, and **higher citric acid**, while showing **lower volatile acidity**. Other features such as residual sugar, free sulfur dioxide, and pH show less separation between classes.

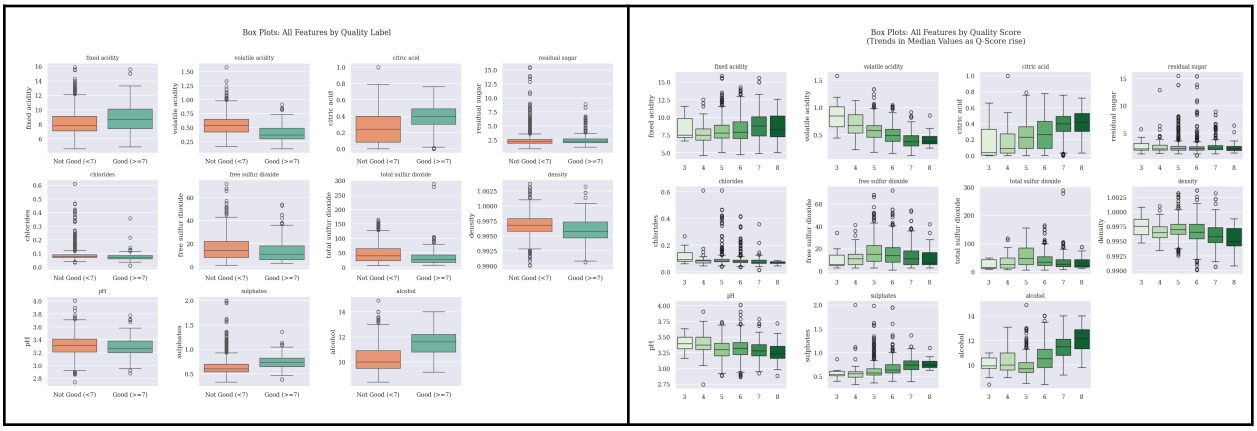


Figure 7: Box plots of all 11 features grouped by quality label and quality scores

4. Regression

4.1 Approach

Linear Regression was applied to predict the continuous quality score (3-8). To assess the influence of feature selection, five different feature combinations were tested, ranging from all 11 features to subsets defined by correlation rank, domain knowledge, and collinearity considerations. All models used standardized features from the same train/test split described in Section 2.

4.2 Feature Combinations

Combo	Name	Features
1	All 11 features	All physicochemical variables
2	Top 3 known predictors	alcohol, volatile acidity, sulphates
3	Flavor profile	alcohol, volatile acidity, citric acid, density
4	Top 5 by correlation	alcohol, volatile acidity, sulphates, citric acid, total sulfur dioxide
5	Low-collinearity set	alcohol, volatile acidity, sulphates, pH, residual sugar

4.3 Results

Combo	Name	Train RMSE	Test RMSE	Train R^2	Test R^2
1	All 11 features	0.6409	0.6687	0.3484	0.3916
2	Top 3 known predictors	0.6557	0.6694	0.3181	0.3903
3	Flavor profile	0.6610	0.6896	0.3069	0.3530
4	Top 5 by correlation	0.6497	0.6756	0.3305	0.3790
5	Low-collinearity set	0.6542	0.6683	0.3211	0.3922

Figure 8: Test RMSE and R^2 for all five regression feature combinations.

4.4 Discussion

All five linear regression models produced somewhat similar performance, with Test RMSE values in the range of 0.669-0.690 and Test R^2 values between 0.353 and 0.392. The best model by Test R^2 was Combo 5 (Low-collinearity set) with R^2 = 0.392, closely followed by Combo 1 (All features) with R^2 = 0.392. Combo 3 (Flavor profile) was the weakest performer.
The consistently moderate R^2 Values across all combinations suggest that linear regression has a fundamental limitation on this task: the relationship between physicochemical features and wine quality is not strongly linear.

Quality scores are also integer-valued and clustered around 5-6, which introduces a ceiling on how well any regression model can fit. Adding more features beyond the top 3 provides only marginal improvement, suggesting that alcohol, volatile acidity, and sulphates carry most of the predictive signal.

5. Classification

5.1 Approach

Binary classification was performed to predict whether a wine is "Good" (quality >= 7) or "Not Good" (quality < 7). Four classifiers were evaluated: Decision Tree, K-Nearest Neighbors (KNN), Gaussian Naive Bayes, and Logistic Regression. Each model was tested on two feature sets: all 11 features and the top 5 features by correlation with quality. Models were compared on accuracy, precision, recall, and F1 score.

5.2 Decision Tree

Decision Tree classifiers were trained with a grid of hyperparameters: max_depth ∈ {3, 5, 10, None} and min_samples_split ∈ {2, 5, 10}, evaluated on both feature sets. The best configuration used all 11 features with max_depth=None and min_samples_split=5.
Result from selected top 6 models:

Max Depth	Min Split	Feature Set	Accuracy	Precision	Recall	F1 Score
None	5	All 11	0.9250	0.7317	0.6977	0.7143
None	2	All 11	0.9250	0.7317	0.6977	0.7143
None	10	All 11	0.9188	0.7180	0.6512	0.6829
10	2	All 11	0.9125	0.6923	0.6279	0.6585
5	5	All 11	0.9063	0.7097	0.5116	0.5946
None	5	Top 5	0.8844	0.5682	0.5814	0.5747

5.3 K-Nearest Neighbors

KNN was evaluated with k ∈ {3, 5, 7, 11} on both feature sets. The best result used the top 5 correlated features with k=3, achieving an F1 score of 0.453. Performance dropped at larger values of k and with all 11 features, suggesting that the additional dimensions introduce noise. Result from selected top 4 models:

K	Feature Set	Accuracy	Precision	Recall	F1 Score
3	Top 5	0.8719	0.5313	0.3953	0.4533
5	Top 5	0.8656	0.5000	0.3256	0.3944
3	All 11	0.8406	0.3889	0.3256	0.3544
7	All 11	0.8719	0.5500	0.2558	0.3492

5.4 Naive Bayes

Gaussian Naive Bayes was applied using both feature sets. The model using all 11 features achieved the better F1 score (0.590) despite lower accuracy than the top-5 model, driven by substantially higher recall (0.721 vs 0.581). Naive Bayes was notably better at capturing the minority "Good" class than KNN or Logistic Regression.

Feature Set	Accuracy	Precision	Recall	F1 Score
All 11 features	0.8656	0.5000	0.7209	0.5905
Top 5 by correlation	0.8844	0.5682	0.5814	0.5747

5.5 Logistic Regression

Logistic Regression (max_iter=1000) was evaluated on both feature sets. Using all 11 features, it achieved accuracy of 0.894 and F1 score of 0.452 - similar to KNN overall, but with much higher accuracy.

Feature Set	Accuracy	Precision	Recall	F1 Score
All 11 features	0.8938	~0.73	~0.33	~0.45
Top 5 by correlation	~0.884	~0.57	~0.46	~0.51

5.6 Model Comparison

Model	Best Feature Set	Accuracy	Precision	Recall	F1 Score
Decision Tree	All 11 features	0.9250	0.7317	0.6977	0.7143
Naive Bayes	All 11 features	0.8656	0.5000	0.7209	0.5905
KNN	Top 5 by correlation	0.8719	0.5313	0.3953	0.4533
Logistic Regression	All 11 features	0.8938	~0.73	~0.33	~0.45

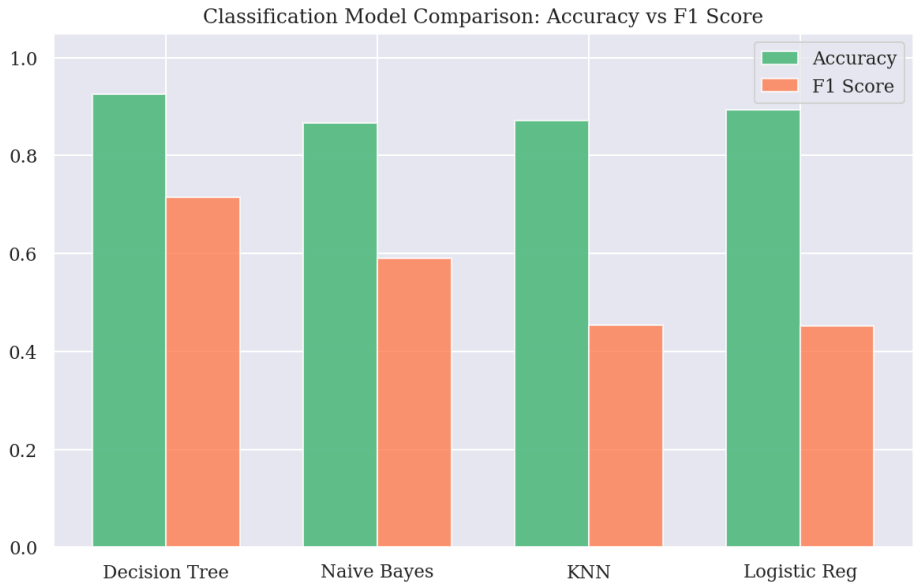
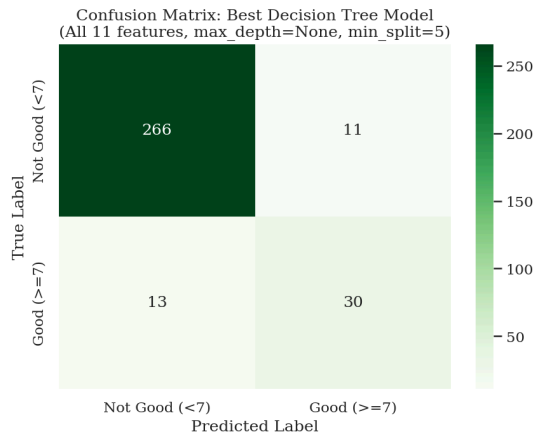


Figure 9: Accuracy and F1 Score comparison across all four classification models.

5.7 Confusion Matrix: Best Model

The best model overall was the Decision Tree classifier (all 11 features, max_depth=None, min_samples_split=5). Its confusion matrix on the 320-sample test set is shown below.

True \ Predicted	Not Good (0)	Good (1)	Confusion Matrix
Not Good (0)	266	11	 Figure 10: Confusion matrix for the best Decision Tree classifier.
Good (1)	13	30	

Out of 320 test samples, the model correctly classified 266 of 277 not-good wines (96% specificity) and 30 of 43 good wines (70% sensitivity). Of the 24 misclassifications, 13 were false negatives (good wines classified as not good) and 11 were false positives.

6. Conclusion

This project applied the full data science pipeline to the task of predicting red wine quality from physicochemical measurements. We evaluated both regression and binary classification approaches across multiple models and feature sets.

6.1 Key Findings

Regression: Linear regression models explained approximately 35-39% of the variance in quality scores ($R^2 \simeq 0.35$-0.39), with RMSE values around 0.67-0.69 on the test set. All five feature combinations produced similar results, indicating that the three strongest features (alcohol, volatile acidity, sulphates) capture most of the linear signal available. The modest R^2 values suggest that wine quality is not well-characterized by a linear combination of these features, and that non-linear methods or additional data may be needed to substantially improve regression performance.

Classification: Binary classification was markedly more successful. The Decision Tree classifier achieved 92.5% accuracy and an F1 score of 71.4% - a strong result given the class imbalance (only 13.6% of wines are "good"). Naive Bayes offered a useful trade-off with higher recall (72.1%), making it a reasonable choice if the cost of missing a good wine is high. KNN and Logistic Regression performed comparably, with F1 scores around 0.45.

Feature importance: Alcohol content was the single most predictive feature across both tasks, followed by volatile acidity and sulphates. Residual sugar, free sulfur dioxide, and pH contributed the least predictive signal.

6.2 Limitations

Several limitations should be noted. First, the dataset is limited to Vinho Verde red wines from a single Portuguese region, so findings may not generalize to other wine types or origins. Second, the strong class imbalance (86% vs 14%) means that even a model with high accuracy may not generalize well to detecting good wines. Third, linear regression is a simple baseline, tree-based ensemble methods (like Random Forest, Gradient Boosting) would likely achieve substantially better regression performance. Finally, the quality labels are themselves derived from human sensory evaluation, introducing subjectivity into the ground truth.

6.3 Future Work

Future directions include applying ensemble classifiers such as Random Forest or XGBoost, which are better suited to the non-linear structure of this problem. Addressing the class imbalance through oversampling techniques (SMOTE) or class-weighted loss functions could further improve recall for the minority class. Combining the red and white wine datasets could also expand the training set and allow cross-type comparisons.